UNIT-V

Cyber Ethics and Laws

Ethical Issues

Computer technology has already taken a key role in the modern era and has become central and essential to major crucial operations in modern society, manufacturing, distribution, transportation, government, defense, education and training, health services, and research and development. The necessity of computers and other handheld devices are likely to increase in the coming days. It has been observed that computer users become vulnerable because of misuse and malfunctions of computers. These create problems of **computer crimes** like hacking, software theft, the spread of viruses, assault of privacy, dependence on intelligent machines, and stress in the workplace. These create one or more ethical crises for computer professionals and the associated professions.

We often blame our computers and such systems for most of our **technological problems** in the present time. When our phone goes down, power supply disrupts, seizing up air traffic control, or traffic light malfunctions, we think computer systems are responsible for these. We even blame computer systems for mistakes by governmental agencies, other utilities, credit-checking bureaus, and even by police forces. Ethical dilemmas that appear before computer users may be summarized as follows:

- Copying software, hacking systems, virus attacks, etc., are the new crises in today's society with existing problems like lack of honesty, trust, confidentiality, responsibility, accountability, and fairness. Almost all of us face a combination of such problems while a computer professional encounter all these nowadays.
- These acts are treated as crimes providing due credits and accountability to digital data and transactions. Such crimes are sometimes regarded as a victimless crime. But these are not victimless in the true sense.
- It is very crucial to provide due access to intellectual property rights of software products and services. These will motivate developers to be creative through awards of necessary incentives.
- A pertinent question is - is hacking always wrong? Of course, the act of ethical hacking offers major activities of tracking malicious activities of cyber criminals.
- Always, we assign the responsibility of system failure to the responsible authorities from whom we purchase the system and warranty for its efficient usages. These must also be defined in case of computer system usages too.
- The information that we store in our file is either private or public depending on the nature of its information character. The same applies to storing data in the computer database also. It is also to be defined when the data is to be offered as information services and its transformation costs.
- The advent of new technologies offered various intelligent systems and smart services. The reliability of these systems is to be characterized so that we can trust such systems.
- The spread of digital technologies may create new health hazards in the workplace. Proper mechanisms need to be defined to handle such hazards. These might also

create the question of whether organizations should allow their higher management to monitor employee activities.

Computer engineers who deal with the design, development, testing, distribution and maintenance of software systems are called **software engineers**. Society and industry expect professional commitment towards analysis, design, development, testing, and maintenance of software systems from software engineers as respected and committed professionals. Software engineers must abide by the following eight principles in accordance with their commitment to the safety, welfare, and health of the public:

- **Public:** Computer engineers must act in the public interest consistently.
- **Client and employer:** Software professionals must act in the best interests of their clients and employer in the public interest.
- **Product**: Software engineers must guarantee that their products, services, and associated modifications meet the best professional standards possible.
- **Judgment**: Software engineers shall maintain integrity and independence in their professional judgment.
- **Management**: Software project managers and leaders have to subscribe and promote an ethical approach to the management of software product development and maintenance.
- **Profession:** Software developers have to maintain the reputation and integrity of the profession in the public interest.
- **Colleagues:** Software professionals have to be fair and helpful to their colleagues.
- **Self:** Computer professionals have to participate in lifelong learning and will promote an ethical approach to the practice of the profession.

Information forms the **intellectual capital** for a person. Where usage and availability of information are concerned, the following ethical issues may arise:

- Intellectual property rights
- Plagiarism
- Digital property rights

Intellectual Property Rights

Intellectual Property (IP) is the property created by an individual or a group of individuals using their own intellect for ultimate use in commerce. This property is not available in the public domain till its official declaration and registration. Examples of intellectual property can be a symbol, a word, or a logo, etc., an invention relating to a product or any process, a new design, a literary or artistic work, or a trademark.

Intellectual Property Rights (IPR) is the statutory right granted by the government to the owner(s) or inventor(s) of the intellectual property. IPR includes copyright, trademarks, industrial design rights, patents, etc. This right is given to the application(s) to prohibit others from exploiting the IP commercially in lieu of the disclosure of his/her IP in the IPR application for a given period of time. It allows IPR owner(s) to make a profit from their invented material(s). Thus it encourages creativity and innovation.

Need of IPR protection

Intellectual property is treated as an asset that can be used by the owner to gain commercially and in any manner he wants. The owner wants to stop others from manufacturing and selling the same or similar products and services that s/he has invented and protected by the IPR. S/he may also like to sell or license the IP to some companies or organizations for commercial gains. Intellectual property can also be used to establish the brand value and goodwill in the market for the inventor and the organizations. Certificates are given to the owner of the intellectual property after a search by the government authority dealing with IPR. It also shows the creators' competence in innovations in the field. This certificate also establishes the valid and legal right about the intellectual property that the inventor(s) has created.

Legal Framework Types of IPRs

IPRs are protected in accordance with the provisions of the legal framework and existing legislations of a country. Following are the acts in India which offer the legal framework for IPR:

- The Copyright Act, 1957.
- The Trade Mark Act, 1999.
- The Geographical Indications of Goods Act, 1999.
- The Patent Act, 1970.
- The Designs Act, 2000.
- Protection of Integrated Circuits Layout and Designs Act, 2000.
- Protection of Plant Varieties and Farmers Rights Act, 2001, and also Trade Secret.

There are various types of IPRs internationally and in India like patents, trade marks, design, geographical indications, and copyright. These are briefly presented in the following table.

Types of IPRs

Sl. No.	IPR Types	Purpose	Corresponding Act
1	Patent	To protect technologies	The Patent Act
2	Trade Mark	To protect words, signs, logos, labels	The Trade Mark Act
3	Design	To protect outer ornamental configuration	The Designs Act
4	Geographical Indications	To protect region-specific product	The Geographical Indications of Goods Act
5	Copyright	To protect literary and artistic work	The Copyright Act

Plagiarism

The word 'plagiarism' was derived from the Latin word 'plagiare'. 'Plagiare' in the Latin language means to abduct or kidnap. Plagiarism is "the act of presenting the words, ideas,

images, sounds, or the creative expression of others as it is your creation or your own". It is very significant to understand the importance of plagiarism with respect to academics and creativity. Plagiarism is the stealing of intellectual property of other creators. It is also considered as cheating and unethical activities. Normally these illegal activities are treated as academic theft and thus come under academic offenses.

There are two types of plagiarism – intentional and unintentional. Copying others' work, cut-paste from electronic resources, borrowing assignments, and downloading text from the internet, and presenting it as our own work are examples of intentional plagiarism. If we do not acknowledge through proper paraphrasing, summarizing, and quotation, then that becomes unintentional plagiarism. Careless copying from electronic databases, quoting excessively, and failure to use our own words are also examples of unintentional plagiarism.

Using our own ideas and developing our own writing skills is the best way to avoid plagiarism. If these are not essentially possible, we must cite the sources. That means, we must acknowledge the sources when someone else's ideas are used. We have to explicitly tell our readers where the ideas are derived from. Rewriting others' ideas in our own words can also solve the problem. While we need to write about a topic from some sources, we have to take careful notes from the sources and write in our own words on the topic.

Digital Rights Management

Digital rights management (DRM) is a **scheme to control copyrighted material** access using technological aids. Here, we should apply digital technologies so that copyrighted materials remain in the read-only mode and copying of the same could be restricted. DRM includes some computer instructions (or codes) that restrict copying or limit the number of accessing digital products or services. Encryption-decryption schemes may also be used to allow access of copyrighted materials like software, e-books, data, digital content, etc. to the intended licensed users. The license should be provided with encryption keys and users who have decryption keys will be able to access the digital materials.

There are many ways to protect our digital materials such as software, content, and other services. We may restrict a user from saving, editing, sharing the digital materials. We may also restrict the printing of some artworks or documents for a limited number of times. It is also possible to disable or restrict the screenshot capture facilities of the computer. If we set an expiry date of our digital products and services, then these will not be opened or accessible after that fixed time. Such setting of the expiry date for the digital materials will be another option for DRM. These restrictions may be imposed based on the IP address or MAC address of our digital devices. Incorporating watermark artworks is another way to establish ownership and identity of the innovation or creator of the digital products and services.

DRM is facing great challenges nowadays. It is a common fact that everybody is not in agreement upon digital rights management. For example, when we pay for music on a specific device or app, we may wish to be able to play it on any other devices or apps that we love. But often DRM does not permit such freedom.

There are a number of benefits of DRM. DRM ensures ownership of the authors of their digital products and services in addition to a guarantee of income streams from their creative works. These will motivate others to be creative or innovative that will definitely elevate human civilization through technological progress. DRM also assists us to secure our digital repositories and keep them private and safe. It offers better ways for licensing agreements and technological distributions. It also teaches digital practitioners about IPR and copyright issues.

Licensing: Creative Commons, GPL and Apache

A license is a legal document that provides **binding (or agreement) guidelines** to the individual(s) or organization(s) as per law regarding the distribution of some products or services. Software is also a product that offers various services and hence, comes under the umbrella of licensing for its distribution. Software licensing provides the right to make multiple copies of the software without violating copyrights to the end-users. It also characterizes the responsibilities of the parties entering into the license agreement. It may enforce restrictions on how the software will be used. The restrictions include terms and conditions on fair use, protections, limitations, warranties, and disclaimers.

There are many **benefits** of software licensing. The right software license can save our money by using unlicensed software against the law. We can avail of round-the-clock maintenance support against the license for which we have paid. We can claim damages according to the agreement in case any such event arises. The license offers a feel-good and confidence while having such copy compared to the situation of using pirated software copies.

There are three systems of software licensing platforms mostly used in the present time, namely, Creative Commons (CC), General Public License (GPL), and Apache. These are presented briefly here.

Creative Commons (CC): It is an internationally active non-profit organization. This organization provides free licenses for the inventors or creators to make use of it when making their work available to the public in advance under certain conditions. When an article is written or a picture is captured, these digital documents or materials are automatically protected by copyright. Such protection restricts others from using these materials in their own way like putting them online. The creator can select how they want others to use the work under this license agreement. Users know what they can and can't do when the work creator publishes his/her work under CC license. Users can use the work if s/he gets permission from the creator only. Interestingly, CC licenses enable works to be used for education purposes by anyone or an organization without financial profits. Hence, teachers and students can freely copy, share and sometimes modify a CC work without having the permission of the creator.

Advantages of CC license include creators' work will be freely available, shareable, and usable online as per permissions applied on creative work. If permission is given, then additional improvement in creative work including open-source code is possible. CC licensing regime also ensures protection of the original copyright of the creator who can modify it within the clauses of the license.
One of the **disadvantages** of a CC license is we cannot revoke a CC license once given. Subsequent usages can be monitored later only in this regime. We can't ask for a license fee or compensation if one makes a profit from our work provided we have not provided a non-commercial license. There is an ambiguity in the copyright of derived works and hence, we must be careful before attaching a CC license to our work.

General Public License (GNU-GPL): It is the most commonly used free software license. GPL was introduced by Richard Stallman in 1989 of the Free Software Foundation for GNU (internationally acclaimed open-source OS Linux user groups) project. GPL license agreement permits software to be freely used, modified, and redistributed by anyone. WordPress is an example of software released under the GPL license. Anyone can download, run, modify, redistribute free copies, and distribute modified versions. GPL is based on copyleft (utilizing copyright law for the right to copy, share, modify and improve creative works of authorship) concept which plays on the word copyright. GPL itself is under the copyright ownership of Stallman's GNU funded 'Free Software Foundation'.
There are many **advantages** of publishing software under GPL. The development of software in new areas is possible by regular feedback from the users. Such open source initiatives help the free development of several other open-source software. The developer's community can volunteer technical support for these open-source software and thus the cost of software maintenance will be reduced. Software bugs can easily be fixed since the number of people working on them is increasing. Developers get attracted to these open source GPL licensed software as they can try and use it without spending much.
The major **disadvantage** of a GPL license is that a commercial product has to be released as open-source if it incorporates GPL licensed products. Thus, if any project uses a GPL licensed component, the whole project has been incorporated in the GPL banner. Because of this reason, major companies set a ban on the use of GPL products. Besides this, many people are not aware of the stringent terms of GPL.

Apache License: Apache Software Foundation (ASF) of Apache Group has introduced the Apache license as a free and open-source software (FOSS) licensing agreement in 1995. This license was initially the same as the old BSD (Berkeley Software Distribution) covered software license. Apache License v2 was a little more radical departure from the initial version of the license.
Apache license allows copy, modification, and **distribution of the covered software** in source or binary forms. It has the provision of exercising the patent right. All modified or unmodified copies are accompanied by a copy of the license. All modifications are to be clearly marked as the work of the modifier. All notices of copyright, trademark, and patent rights are to be reproduced accurately in distributed copies under this license.

Privacy and Privacy Laws

Privacy is a fundamental right according to our law. This is highly essential to the autonomy and the protection of human dignity. This serves as the foundation for many other human rights. Privacy deals with the ability of an individual or organization to determine what data in a computer system can be shared with third parties with respect to information technology. Privacy is very important with respect to an individual and organizations. It offers self-respect to an individual and also to an organization. Privacy helps us to maintain social boundaries. It enables freedom of thought and speech, and freedom of social and political activities of the person to whom the data belongs. Privacy empowers the company to whom the data belongs, to protect privacy.

Following are the **threats** to the privacy of an individual and organizations in today's world.

- Data collection
- Web tracking
- Lack of security
- Public Wi-Fi
- Social networking
- Connected everything
- Government spying

Privacy laws are regulations that protect a person's or organization's data confidentiality. It governs the collection, storage, and release of financial, medical, and other personal information to third parties. Privacy laws are of several types - general privacy laws (overall bearing on the personal information of individual), and specific privacy laws (to regulate specific types of information) such as communication privacy laws, information privacy laws, health privacy laws, financial privacy laws, online privacy laws.

IT-Act 2000, explained later, deals with the problems relating to the payment of compensation (civil) and punishment (criminal) in case of unlawful disclosure and exploitation of personal data and infringement of contractual terms in respect of personal data.

Under section 43A of IT-Act, 2000, a corporate body may be held responsible for possessing, dealing, or handling any sensitive personal information. The rules only deal with the protection of sensitive personal data or information of a person, which includes such personal information which consists of information relating to:

Financial information (such as bank account or credit card or debit card or other payment instrument details)
Passwords
Physiological information
Physical health conditions
Sexual orientation
Biometric information
Medical records and history

Under section 72A of the IT-Act, 2000, exposé of information, knowingly and intentionally, without the consent of the person concerned and breach of lawful contract is punishable with imprisonment for a term extending up to 3 years and fine extending up to □5L.

IT Act-2000

Information Technology Act, 2000 or IT Act-2000 is the act of the Indian Parliament notified on 17th October, 2000 to provide a **legal framework** for electronic governance. It provides **recognition to electronic records, digital signatures, electronic exchange of data, and electronic commerce transactions**. It is the primary law dealing with electronic commerce and cybercrimes in India. The original act contained 94 sections, divided into 4 schedules and 19 chapters. This law is inherited from the United Nations Commission for International Trade related Laws (UNCITRAL) model law. The laws, under this act, are applicable to the whole of India. It is applicable to persons of other nationalities if the crime involves a computer or network located in India.

We can divide the commission of cybercrimes into the following **three groups**:

Types of cybercrimes

Against Individual	Against Organisation	Against Society at Large
Hacking/cracking Harassment via emails Cyber stalking Dissemination of obscene material Defamation Hacking/cracking indecent exposure Computer vandalism Transmitting a virus Network trespassing Unauthorized control over computer system	Hacking & cracking Possession of unauthorized information Cyber- terrorism against a government organization Distribution of pirated software etc.	Pornography Polluting the youth through indecent exposure Trafficking

Some of the **important sections** under the IT Act-2000 are given below:

Important offenses and penalties under the IT Act-2000

Sections	Offense	Penalty
67A	Publishing images containing sexual acts.	Imprisonment up to 7 years, or/and with fine up to □10L.
67B	Publishing child porn or predating children online.	Imprisonment up to 5 years, or/and with fine up to □10L on the first conviction. Imprisonment up to 7 years, or/and with fine up to □10L on second conviction.
68	Failure to maintain records.	Imprisonment up to 3 years, or/and with a fine.
69	Failure/refusal to comply	Imprisonment up to 3 years, or/and with fine up to

	with orders.	□2L.
70	Attempting to secure access to a protected system.	Imprisonment up to 10 years, or/and with a fine.
71	Misrepresentation	Imprisonment up to 3 years, or/and with fine up to □1L.

IT Act, 2000 was **amended** in December, 2008. Indian parliament passed the act IT (Amendment) Act, 2008 (IT Act 2008) to provide more focus on information security. Cyber terrorism and data protection were incorporated as several additional sections. It came into force on 27th October, 2009. Following are some major amendments in this act:

Digital signature gets legal recognition for authentication of digital documents.

Legal recognition was awarded to digital documents. Government documents may also be supplied in electronic form.

A maximum fine of up to □1 Crore may be imposed for any damage to computer and computer systems.

Other relevant acts (such as Indian Penal Code – 1860, Indian Evidence Act – 1872, Bankers' Book Evidence Act – 1891, Reserve Bank of India Act - 1934) of IT Act 2008 were also amended to align them with the act.

E-Commerce

E-Commerce or Electronic Commerce means buying and selling of goods, products, or services over the internet. E-commerce is also known as electronic commerce or internet commerce. These services provided online over the internet network. Transaction of money, funds, and data are also considered as E-commerce. These business transactions can be done in four ways: Business to Business (B2B), Business to Customer (B2C), Customer to Customer (C2C), Customer to Business (C2B). The standard definition of E-commerce is a commercial transaction which is happened over the internet. Online stores like Amazon, Flipkart, Shopify, Myntra, Ebay, Quikr, Olx are examples of E-commerce websites. By 2020, global retail e-commerce can reach up to $27 Trillion. Let us learn in detail about what is the advantages and disadvantages of E-commerce and its types.

E-Commerce or Electronic Commerce

E-commerce is a popular term for electronic commerce or even internet commerce. The name is self-explanatory, it is the meeting of buyers and sellers on the internet. This involves the transaction of goods and services, the transfer of funds and the exchange of data.

So when you log into your Amazon and purchase a book, this is a classic example of an e-commerce transaction. Here you interact with the seller (Amazon), exchange data in form of pictures, text, address for delivery etc. and then you make the payment.

As of now, e-commerce is one of the fastest growing industries in the global economy. As per one estimate, it grows nearly 23% every year. And it is projected to be a $27 trillion industry by the end of this decade.

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- Digital Economy
- M-Commerce

Types of E-Commerce Models

Electronic commerce can be classified into four main categories. The basis for this simple classification is the parties that are involved in the transactions. So the four basic electronic commerce models are as follows,

1. Business to Business

This is Business to Business transactions. Here the companies are doing business with each other. The final consumer is not involved. So the online transactions only involve the manufacturers, wholesalers, retailers etc.

2. Business to Consumer

Business to Consumer. Here the company will sell their goods and/or services directly to the consumer. The consumer can browse their websites and look at products, pictures, read reviews. Then they place their order and the company ships the goods directly to them. Popular examples are Amazon, Flipkart, Jabong etc.

3. Consumer to Consumer

Consumer to consumer, where the consumers are in direct contact with each other. No company is involved. It helps people sell their personal goods and assets directly to an interested party. Usually, goods traded are cars, bikes, electronics etc. OLX, Quikr etc follow this model.

4. Consumer to Business

This is the reverse of B2C, it is a consumer to business. So the consumer provides a good or some service to the company. Say for example an IT freelancer who demos and sells his software to a company. This would be a C2B transaction.

Examples of E-Commerce

- Amazon
- Flipkart
- eBay
- Fiverr
- Upwork
- Olx
- Quikr

Advantages of E-Commerce

- E-commerce provides the sellers with a global reach. They remove the barrier of place (geography). Now sellers and buyers can meet in the virtual world, without the hindrance of location.
- Electronic commerce will substantially lower the transaction cost. It eliminates many fixed costs of maintaining brick and mortar shops. This allows the companies to enjoy a much higher margin of profit.
- It provides quick delivery of goods with very little effort on part of the customer. Customer complaints are also addressed quickly. It also saves time, energy and effort for both the consumers and the company.
- One other great advantage is the convenience it offers. A customer can shop 24×7. The website is functional at all times, it does not have working hours like a shop.
- Electronic commerce also allows the customer and the business to be in touch directly, without any intermediaries. This allows for quick communication and transactions. It also gives a valuable personal touch.

Disadvantages of E-Commerce

- The start-up costs of the e-commerce portal are very high. The setup of the hardware and the software, the training cost of employees, the constant maintenance and upkeep are all quite expensive.
- Although it may seem like a sure thing, the e-commerce industry has a high risk of failure. Many companies riding the dot-com wave of the 2000s have failed miserably. The high risk of failure remains even today.
- At times, e-commerce can feel impersonal. So it lacks the warmth of an interpersonal relationship which is important for many brands and products. This lack of a personal touch can be a disadvantage for many types of services and products like interior designing or the jewelry business.
- Security is another area of concern. Only recently, we have witnessed many security breaches where the information of the customers was stolen. Credit card theft, identity theft etc. remain big concerns with the customers.
- Then there are also fulfillment problems. Even after the order is placed there can be problems with shipping, delivery, mix-ups etc. This leaves the customers unhappy and dissatisfied.

E- Commerce and E-Governance

What is commerce?
Commerce is an important part of a business. It is nothing but buying and selling of goods, which means when we buy a product or service to others, and then it is called as commerce.

What is E-commerce?
E-commerce can be broadly defined as the process of buying and selling of goods or services using an electronic medium such as the Internet. It is also referred as a paperless exchange of business information using EDI, E-mail, electronic fund transfer, etc.

Differences between Traditional commerce and E-commerce

Traditional commerce	**E-commerce**
It focuses on the exchange of products and services through personal interactions, so it is manual.	Trading activities are online via the internet and can be considered automatic.
Traditional commerce is limited to business hours i.e. during the day	It is 24×7; it can be done anytime day and night.
Traditional commerce provides face to face customer interaction.	It can be termed as screen to face interaction.
It is limited to a particular geographic location.	It is global and has no physical limitation.
Modes of payment: cash, cheques and credit cards.	Modes of payment: bank transfer, credit cards, e-wallet, mobile payment and many more.
Delivery of goods or services is instant.	Delivery of goods or services takes some time.
Its scope is local.	Its scope is global.

Advantages of E-Commerce

- **Global scope**: It provides the sellers with a global reach. Now sellers and buyers can meet in the virtual world, without a geographical barrier.
- **Electronic transaction**: E-commerce reduces the paper work and significantly lowers the transaction cost. It enables the use of credit cards, debit cards, smart cards, electronic fund transfer via bank's website and other modes of electronic payment.
- **Cost saving**: E-commerce application provides users with more options to compare and select the cheaper and better option. It helps in reducing services such as healthcare, the cost of searching a product, etc. E-commerce has enabled rural areas to access services and products, which are otherwise not available to them.
- **Anytime shopping**: One other great advantage is the convenience. A customer can shop 24×7. The website is functional at all times; it does not have working hours like a shop.

- **No intermediaries**: Electronic commerce also allows the customer and the business to be in touch directly, without any intermediaries. This allows for quick communication and transactions.
- **Public services**: E-commerce helps the government to deliver public e education, social services at a reduced cost and in an improved manner.

Disadvantages of E-Commerce

- **Setup cost:** The setup of the hardware and the software, the training cost of employees, the constant maintenance and upkeep are all quite expensive
- **Physical presence:** This lack of a personal touch can be a disadvantage for many types of services and products like interior designing or the jewellery business.
- **Security:** Security is another area of concern. Credit card theft, identity theft etc. remain big concerns with the customers.
- **Goods delivery:** There may arrive some problem with fulfilment of order. Even after the order is placed there can be problems with shipping, delivery, mix-ups etc. This leaves the customers unhappy and dissatisfied

Types of E-Commerce

- **Business – to – Consumer (B2C**): In B2C model, business sells its products directly to a customer. A customer can view the products shown on the website. The customer can choose a product and order the same. The website will then send a notification to the business organization via email and the organization will dispatch the product/ goods to the customer. These B2C businesses are online retailers. Example: Amazon, Flipchart, etc.
- **Business – to – Business (B2B):** In B2B model, business sells its products to an intermediate buyer who then sells the product to the final customer. As an example, a wholesaler places an order from a company's website and after receiving the consignment, sells the product to the final customer who comes to buy the product at one of its retail outlets. Example: Tata communications (network provider).
- **Consumer – to – Consumer (C2C**): In C2C model, consumer helps consumer to sell their assets like residential property, cars, motorcycles , or rent a room by publishing their information on the website. Website may or may not charge the consumer for its services. Example OLX, Quirk, online auction.
- **Consumer – to – Business (C2B):** In this model, consumers have products or services of value that can be consumed by businesses. For example, the comparison of interest rates of personal loan/car loan provided by various banks via websites. A business organization that fulfills the consumer's requirement within the specified budget approaches the customer and provides its services. For e.g. – A blog can be written by an author for a business to improve sale of products, eBay.

Phases of a trade cycle

- Pre-sales
- Execution
- Settlement
- After-sales

Different modes of payment

- Credit cards
- Mobile payments
- Bank transfers
- E-wallets

What is M-commerce?

- It is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets.
- It enables users to access online shopping platforms without needing to use a desktop computer. Some of application of M-Commerce are mobile banking, ticket booking, E-bill payment, online auctions, stock market trading.

Use of social commerce

- Social commerce is a form of electronic commerce that involves social media, online media that supports social interaction.
- It enables shoppers to get advice from trusted individuals, find goods and services and then purchase them.
- The social networks that spread this advice have been found to increase the customer's trust in one retailer over another.
- It uses networking websites such as Facebook, Instagram and Twitter as vehicles to promote and sell products and services.
- The success of a social commerce campaign is measured by the degree to which consumers interact with the company's marketing through retweets, likes and shares.

E-Commerce technology- Electronic Data Interchange (EDI)

- EDI is the electronic interchange of business information using a standardized format; a process which allows one company to send information to another company electronically rather than on paper.
- Business entities conducting business electronically are called trading partners.
- Many business documents can be exchanged using EDI; two most common documents are purchase orders and invoices.
- It is computer-to-computer interchange of strictly formatted documents via telecommunications or physically transported on electronic storage media.

What is E-Governance?

- It signifies the implementation of information technology in the government processes and functions so as to cause simple, moral, accountable and transparent governance.
- The basic purpose of e-governance is to simplify processes for all, i.e. government, citizens, businesses etc. at National, State and local levels.
- Hence, E-governance delivers SMART government. (S- Simple, M-Moral, A-Accessible, R-Responsive, T-Transparent Government)

Advantages of E-Governance

- Reduced corruption
- High transparency
- Increased convenience
- Direct participation of constituents

- Reduction in overall cost.
- Expanded reach of government

Types of E-Governance

- Government-to-Citizen (G2C)
- Government-to-Business (G2B)
- Government-to-Government (G2G)
- Government-to-Employee (G2E)

Government-to-Citizen E-Governance

- This refers to the government services which enable citizens to get access to wide variety of public services.
- Most of the government services fall under G2C. It helps the ordinary people reduce the time and cost to conduct a transaction.
- A citizen can have access to the services anytime from anywhere.
- Many services like license renewals and paying tax are essential in G2C. It also focuses on geographic land barriers.

Government-to-Business (G2B) type E-Governance

- The Government to business is the exchange of services between Government and Business organizations.
- G2B provides access to relevant forms needed to comply.
- The G2B also consists of many services exchanged between business sectors and government.
- It aims at eliminating paper work, saving time, cost and establish transparency in the business environment, while interacting with government.

Government-to-Government (G2G)

- The Government-to-Government refers to the interaction between different government departments, organizations and agencies.
- In G2G, government agencies can share the same database using online communication. The government departments can work together.
- G2G services can be at the local level or the international level. Likewise, it provides safe and secure inter-relationship between domestic or foreign government.

Government-to-Employee (G2E)

- The Government-to-Employee is the internal part of G2G sector. G2E aims to bring employees together and improvise knowledge sharing.
- It provides online facilities to the employees like applying for leave, reviewing salary payment record and checking the balance of holiday.
- The G2E sector provides human resource training and development.
- G2E is also the relationship between employees, government institutions and their management.

Some effective examples of successful implementation of E-Governance projects

- e-Mitra project (Rajasthan)
- e-Seva project (Andhra Pradesh)
- CET (Common Entrance Test)
- AADHAAR card

- DigiLocker
- Bharat Bill Payment System
- PAN
- EPFO services
- PMKVY services

Security measures in E-Commerce.

- Encryption
- Digital signature
- Digital certificate

What is encryption?

Encryption is widely used on the internet to protect user information being sent between a browser and a server. This includes passwords, payment information and other personal information that should be considered private.

What are the types of encryption?

- Symmetric (Shared Secret Encryption)
- Asymmetric (Public-Key Encryption)

What is the use of digital signatures?

A digital signature is also known as an electronic signature. A digital signature guarantees the authenticity of an electronic document or message in digital communication and uses encryption technique (asymmetric cryptography) to provide proof of original and unmodified documentation. Digital signatures are used in e-commerce, software distribution, and financial transactions. This is the direct transfer of information between two partners.

What is a digital certificate?

A Digital Certificate is an electronic "password" that allows a person or organization to exchange data securely over the Internet using the public key infrastructure (PKI). Digital Certificate is also known as a public key certificate or identity certificate. In this, information is transferred between two authorized partners who have digital certificates issued by some supreme authority.

Network Layer-IPSec

IP security (IPSec)

IP Sec (Internet Protocol Security) is an Internet Engineering Task Force (IETF) standard suite of protocols between two communication points across the IP network that provide data authentication, integrity, and confidentiality. It also defines the encrypted, decrypted, and authenticated packets. The protocols needed for secure key exchange and key management are defined in it.

Uses of IP Security

IPsec can be used to do the following things:

- To encrypt application layer data.
- To provide security for routers sending routing data across the public internet.
- To provide authentication without encryption, like to authenticate that the data originates from a known sender.

- To protect network data by setting up circuits using IPsec tunneling in which all data being sent between the two endpoints is encrypted, as with a Virtual Private Network(VPN) connection.

Components of IP Security
It has the following components:

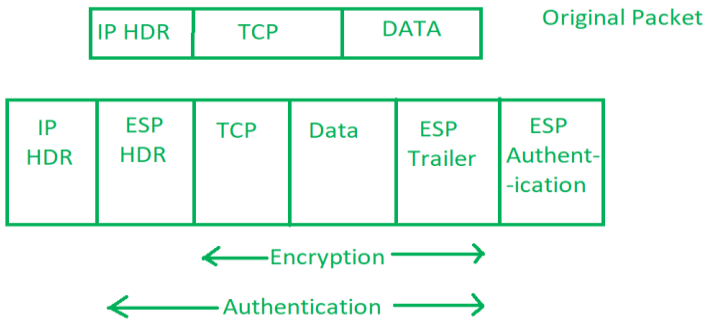
1. Encapsulating Security Payload (ESP)
2. Authentication Header (AH)
3. Internet Key Exchange (IKE)

1. Encapsulating Security Payload (ESP): It provides data integrity, encryption, authentication, and anti-replay. It also provides authentication for payload.
2. Authentication Header (AH): It also provides data integrity, authentication, and anti-replay and it does not provide encryption. The anti-replay protection protects against the unauthorized transmission of packets. It does not protect data confidentiality.

IP HDR	AH	TCP	DATA

IP Header

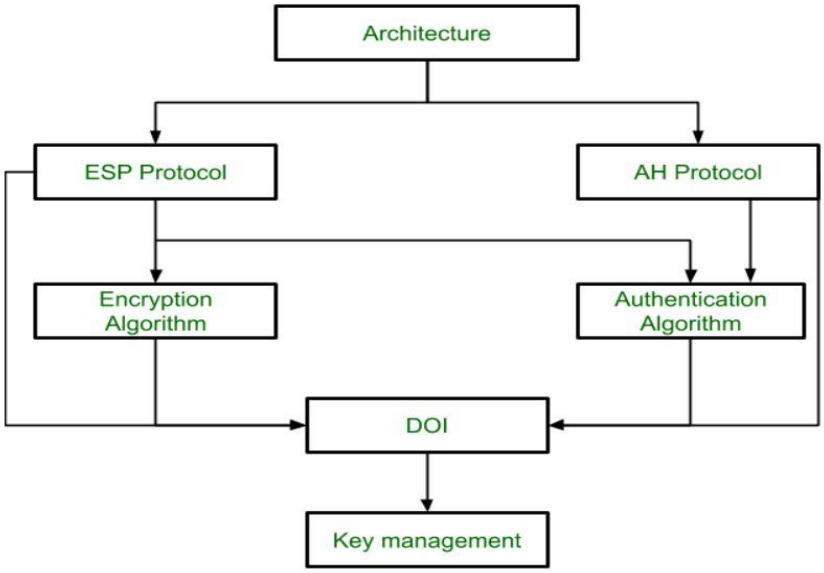
3. Internet Key Exchange (IKE): It is a network security protocol designed to dynamically exchange encryption keys and find a way over Security Association (SA) between 2 devices. The Security Association (SA) establishes shared security attributes between 2 network entities to support secure communication. The Key Management Protocol (ISAKMP) and Internet Security Association provide a framework for authentication and key exchange. ISAKMP tells how the setup of the Security Associations (SAs) and how direct connections between two hosts are using IPsec. Internet Key Exchange (IKE) provides message content protection and also an open frame for implementing standard algorithms such as SHA and MD5. The algorithm's IP sec users produce a unique identifier for each packet. This identifier then allows a device to determine whether a packet has been correct or not. Packets that are not authorized are discarded and not given to the receiver.



Packets in Internet Protocol

IP Security Architecture
IPSec (IP Security) architecture uses two protocols to secure the traffic or data flow. These protocols are ESP (Encapsulation Security Payload) and AH (Authentication Header). IPSec Architecture includes protocols, algorithms, DOI, and Key Management. All these components are very important in order to provide the three main services:

- Confidentiality
- Authenticity
- Integrity



IP Security Architecture

How does IPsec work?

IPsec connections include the following steps:

Key exchange: Keys are necessary for encryption; a key is a string of random characters that can be used to "lock" (encrypt) and "unlock" (decrypt) messages. IPsec sets up keys with a key exchange between the connected devices, so that each device can decrypt the other device's messages.

Packet headers and trailers: All data that is sent over a network is broken down into smaller pieces called packets. Packets contain both a payload, or the actual data being sent, and headers, or information about that data so that computers receiving the packets know what to do with them. IPsec adds several headers to data packets containing authentication and encryption information. IPsec also adds trailers, which go after each packet's payload instead of before.

Authentication: IPsec provides authentication for each packet, like a stamp of authenticity on a collectible item. This ensures that packets are from a trusted source and not an attacker.

Encryption: IPsec encrypts the payloads within each packet and each packet's IP header (unless transport mode is used instead of tunnel mode). This keeps data sent over IPsec secure and private.

Transmission: Encrypted IPsec packets travel across one or more networks to their destination using a transport protocol. At this stage, IPsec traffic differs from regular IP traffic in that it most often uses UDP as its transport protocol, rather than TCP. TCP, the Transmission Control Protocol, sets up dedicated connections between devices and ensures that all packets arrive. UDP, the User Datagram Protocol, does not set up these dedicated connections. IPsec uses UDP because this allows IPsec packets to get through firewalls.

Decryption: At the other end of the communication, the packets are decrypted, and applications (e.g. a browser) can now use the delivered data.

Features of IPSec

1. **Authentication:** IPSec provides authentication of IP packets using digital signatures or shared secrets. This helps ensure that the packets are not tampered with or forged.
2. **Confidentiality:** IPSec provides confidentiality by encrypting IP packets, preventing eavesdropping on the network traffic.
3. **Integrity:** IPSec provides integrity by ensuring that IP packets have not been modified or corrupted during transmission.
4. **Key management:** IPSec provides key management services, including key exchange and key revocation, to ensure that cryptographic keys are securely managed.
5. **Tunneling:** IPSec supports tunneling, allowing IP packets to be encapsulated within another protocol, such as GRE (Generic Routing Encapsulation) or L2TP (Layer 2 Tunneling Protocol).
6. **Flexibility:** IPSec can be configured to provide security for a wide range of network topologies, including point-to-point, site-to-site, and remote access connections.
7. **Interoperability:** IPSec is an open standard protocol, which means that it is supported by a wide range of vendors and can be used in heterogeneous environments.

Advantages of IPSec

1. **Strong security:** IPSec provides strong cryptographic security services that help protect sensitive data and ensure network privacy and integrity.
2. **Wide compatibility:** IPSec is an open standard protocol that is widely supported by vendors and can be used in heterogeneous environments.
3. **Flexibility:** IPSec can be configured to provide security for a wide range of network topologies, including point-to-point, site-to-site, and remote access connections.
4. **Scalability:** IPSec can be used to secure large-scale networks and can be scaled up or down as needed.
5. **Improved network performance:** IPSec can help improve network performance by reducing network congestion and improving network efficiency.

Disadvantages of IPSec

1. **Configuration complexity:** IPSec can be complex to configure and requires specialized knowledge and skills.
2. **Compatibility issues:** IPSec can have compatibility issues with some network devices and applications, which can lead to interoperability problems.
3. **Performance impact:** IPSec can impact network performance due to the overhead of encryption and decryption of IP packets.
4. **Key management:** IPSec requires effective key management to ensure the security of the cryptographic keys used for encryption and authentication.
5. **Limited protection:** IPSec only provides protection for IP traffic, and other protocols such as ICMP, DNS, and routing protocols may still be vulnerable to attacks.